

Cosmic muons through the ESA test stand

A rate estimate with and without 3 ft of concrete

18 September 2026

CAD-registered geometry: eae30ac8f

Result. The current geometry gives approximately **1.41 muons/hour without an overhead slab** and **1.20 muons/hour with 3 ft of concrete**. Both numbers require a single muon to intersect all 26 active layers. They are conditional predictions from a sea-level spectrum and a straight-line continuous-slowng-down model, before detector efficiency; they are not measured rates or a full Geant4 transport result.

Overhead shielding	Muons/hour	Muons/day	Mean interval	$K_{\min}$ at surface
None	1.41	33.8	42.6 min	0.104 GeV
3 ft concrete	1.20	28.8	50.1 min	0.470 GeV

Here $K_{\min}$ is the acceptance-weighted mean kinetic energy needed to emerge from the modelled detector with at least 10 MeV remaining. That small residual is a range-table reliability floor, not an electronics threshold. The concrete reduces this model's rate by approximately **15%**.

What counts as going through everything?

A trajectory must have nonzero path length in an active volume in *every layer* listed below. Within a segmented layer, hitting any one bar is sufficient. A track need not hit all bars in that layer.

Subsystem	Required layers	Active volumes	Geometry used
Top HCal	2	24	Two crossed layers, 12 bars each
Trigger counters	6	72	Three stations, two staggered layers each
Silicon tracker	4	4	Active daughters; inactive borders excluded
LYSO target	2	33	Two staggered bar arrays
HCal above ECal	4	32	Four crossed layers, 8 bars each
ECal silicon	4	4	Four hexagonal active silicon layers
HCal below ECal	4	48	Four crossed layers, 12 bars each
Total	26	217	Bars and sensors are distinct volumes

The STEP model supplies the placement framework; ESA **ldmx-reduced-v3** supplies the ECal, trigger, tracker and target internals. HCal dimensions follow the test-stand branch. The inferred HCal bar counts and detector registration still need hardware confirmation. The support renderings are not additional material in the GDML used here.

The geometric bottleneck

The trigger bars are only 30 mm wide in X , while the HCal spans metres. After requiring the six trigger and two LYSO layers, the tracker and HCal do not remove further tracks in this integration; the displaced ECal footprint reduces the remaining rate by about 36%. The accepted rays are close to vertical: 90% are within 4.82° and 99% within 6.28° under the reference angular model.

For the simple reference intensity $I(\theta) = 70 \cos^2 \theta \text{ m}^{-2}\text{s}^{-1}\text{sr}^{-1}$, associated with sea-level muons above 1 GeV/ c [1],

$$G_2 = \int dA d\Omega \cos^3 \theta \mathbf{1}_{\text{all}} = 4.0021 \times 10^{-6} \text{ m}^2\text{sr}, \quad R_{p>1}^{\text{geom}} = 1.0085 \pm 0.0004 \text{ hour}^{-1}.$$

The quoted error is numerical integration error only. The headline rates are larger because the energy-dependent calculation also includes lower-energy muons that have enough range to cross the selected materials.

Geometry and shielding illustrations

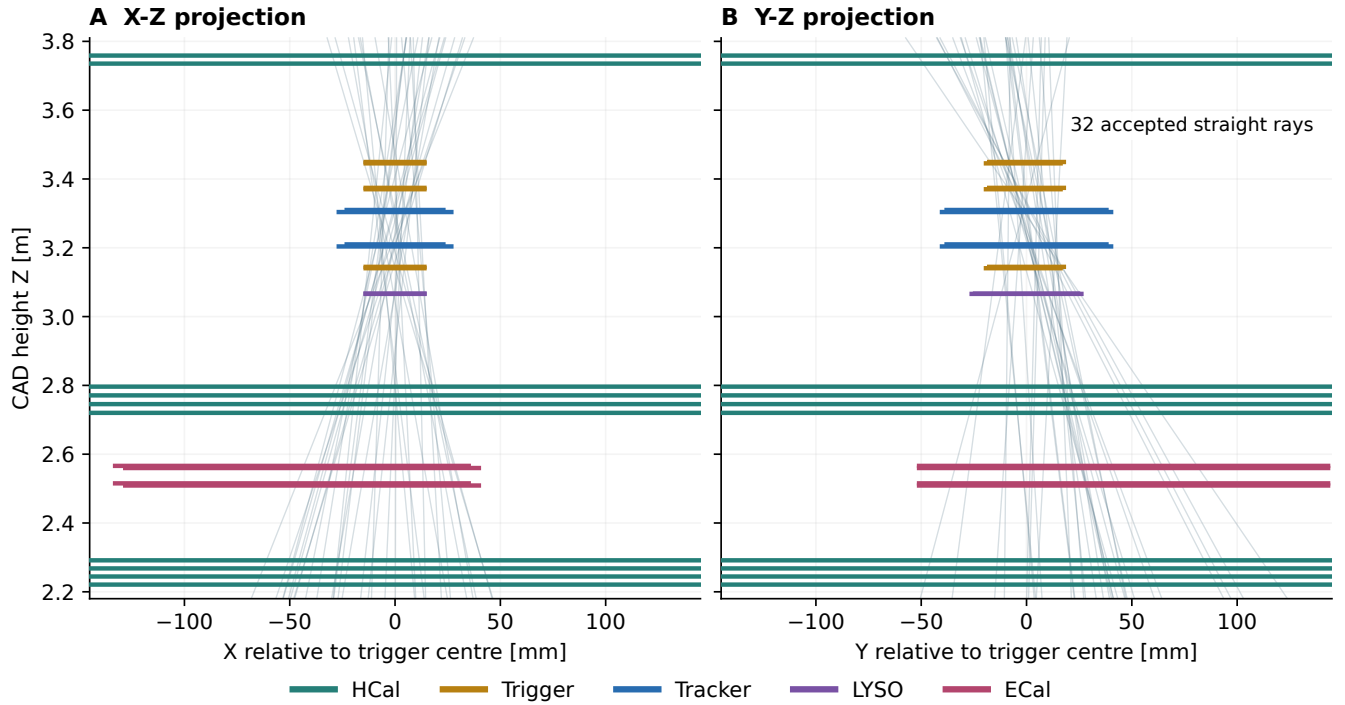


Figure 1. Two projections of 32 accepted sample trajectories. Coloured segments mark the actual layer centres and transverse extents; their line thickness is enlarged for legibility. HCal bars extend beyond the displayed transverse window. The ECal offset and narrow trigger/LYSO corridor restrict the accepted directions. These are straight geometric rays, not simulated scattering histories.

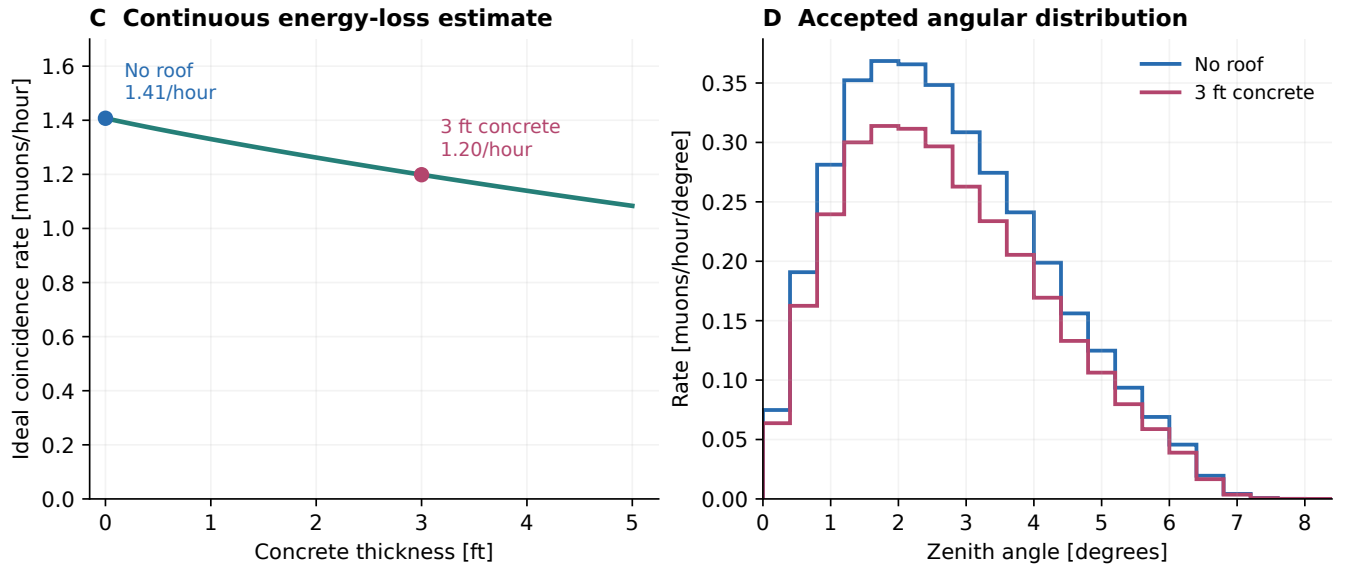


Figure 2. The continuous-loss estimate versus concrete thickness, and the angular distribution of muons that satisfy all 26 layer requirements. The 3 ft comparison assumes a uniform horizontal slab that covers the stand and all accepted directions, with concrete density 2.30 g cm^{-3} [3]. Roof height and openings are not modelled. No efficiency correction or calibrated uncertainty band is implied.

Calculation and checks

Geometry integration. Two points are sampled on horizontal planes through the upper trigger and lower LYSO layer, separated by $D = 383.4$ mm. Sampling rectangles are slightly expanded to include tracks that clip a finite-thickness anchor volume. If $\mathbf{u} = (\mathbf{b} - \mathbf{a})/D$, then

$$R = \int d^2a d^2b \frac{I(\theta) \cos^4 \theta}{D^2} \mathbf{1}_{\text{all}}, \quad \cos \theta = (1 + |\mathbf{u}|^2)^{-1/2}.$$

This includes both the projected-area factor and the solid-angle Jacobian. For $I = I_0 \cos^2 \theta$, each ray carries a $\cos^6 \theta$ weight. Box and hexagonal-prism intersections retain the physical bar gaps, stagger, tracker rotations and finite layer thickness. Eight independently scrambled Sobol sequences of 2^{18} point pairs give 2,097,152 trials. An independent pseudorandom estimate agrees within its numerical error. ROOT's original GDML solids agree with all 4,992 individual layer decisions in a separate 192-ray test.

Spectrum. The energy-dependent model uses Guan et al.'s modified Gaisser spectrum[2], with total energy E in GeV and j in $\text{cm}^{-2}\text{s}^{-1}\text{sr}^{-1}\text{GeV}^{-1}$:

$$j(E, \theta) = C 0.14 \left[E + \frac{3.64}{(\cos \theta^*)^{1.29}} \right]^{-2.7} \left[\frac{1}{1 + 1.1E \cos \theta^*/115} + \frac{0.054}{1 + 1.1E \cos \theta^*/850} \right].$$

The factor $C = 1.1595$ sets the vertical integral above $p = 1$ GeV/ c to $70 \text{ m}^{-2}\text{s}^{-1}\text{sr}^{-1}$. The published curvature correction for θ^* is implemented in the supplied script. The flux above each ray's required surface energy replaces the reference $70 \cos^2 \theta$ weight. The resulting lower-energy contribution is model dependent.

Stopping model. The supplied 3 ft thickness is 91.44 cm, or $210.312 \text{ g cm}^{-2} = 2.10312 \text{ m}$ water equivalent at the assumed density. ROOT chords and the GDML densities give a mean vertical detector column of 32.70 g cm^{-2} over 96 accepted rays, of which roughly 20.6 g cm^{-2} is HCal scintillator. For this estimate, the whole detector column is represented by a polystyrene CSDA range[4]; the roof uses the concrete CSDA table[3]. This is an approximation to the mixed detector materials, not material-by-material transport. With tabulated range functions $\mathcal{R}_p$ and $\mathcal{R}_c$,

$$K_d = \mathcal{R}_p^{-1}[\mathcal{R}_p(10 \text{ MeV}) + X_d \sec \theta], \quad K_s = \mathcal{R}_c^{-1}[\mathcal{R}_c(K_d) + X_c \sec \theta].$$

The flux integral starts at $E = K_s + m_\mu$. With no roof, $K_s = K_d$. Doubling the detector column changes the 3 ft prediction from 1.20 to 1.17/hour; requiring 100 MeV at the final exit gives 1.17/hour.

Interpretation. The recorded coincidence rate will also depend on the light-yield thresholds, dead channels, trigger logic and live time. Multiple scattering, decay and stochastic energy losses are omitted. Grazing tracks can count here: requiring a hit at every layer's midplane instead reduces the reference rate from 1.0085 to 0.853/hour, showing that the signal threshold matters. The flux normalization itself is not exact; the PDG review notes measurements 10–15% below the reference value[1]. No combined systematic uncertainty or local flux calibration is claimed. The estimate is for cosmic muons of both charges, not all cosmic-ray species or shower coincidences.

Reproduction. The folder contains the geometry snapshot, scripts, numerical tables, range data, figures and this L^AT_EX source. Run `build_report.sh` with Python (NumPy, SciPy, Matplotlib), ROOT and Tectonic available. The SHA256SUMS file records artifact identities. The original STEP SHA-256 and all source GDML hashes are preserved in `geometry/provenance.json`.

References

- [1] Particle Data Group, *Cosmic Rays*, 2022 review, Sec. 30.3.1. <https://pdg.lbl.gov/2022/reviews/rpp2022-rev-cosmic-rays.pdf>.
- [2] M. Guan et al., *A parametrization of the cosmic-ray muon flux at sea-level*, arXiv:1509.06176 (2015), Eqs. (2)–(3). <https://arxiv.org/abs/1509.06176>.
- [3] PDG, *Shielding concrete: properties and muon CSDA range*. https://pdg.lbl.gov/2022/AtomicNuclearProperties/HTML/shielding_concrete.html.
- [4] PDG, *Muon stopping power and range in polystyrene*. https://pdg.lbl.gov/2019/AtomicNuclearProperties/MUE/muE_polystyrene.txt.

Raising the TS/tracker/LYSO assembly by 0.20 m

All three trigger-scintillator (TS) stations, all four tracker sensors and both LYSO layers move together by +200 mm in CAD Z. Their transverse positions, rotations, internal spacings and segmentation stay the same. ECal and all three HCal groups remain fixed. A muon must still cross every one of the 26 active layers.

Raising this assembly alone reduces the estimated rate by 10.2%. The top-TS to lower-LYSO separation stays at 383.4 mm, preserving the internal collimation, but the distance to the fixed, offset ECal grows by 200 mm. Fewer accepted directions also cross every ECal layer. The geometric $p > 1$ GeV/ c reference changes from 1.0085 to 0.9059 ± 0.0003 muons/hour; the error is numerical only.

Overhead shielding	Original/hour	Raised/hour	Change
None	1.407	1.264	−10.2%
3 ft concrete	1.199	1.077	−10.2%

Centering the ECal as well

Keep the TS/tracker/LYSO assembly raised and translate the entire ECal in X, Y so that the centre of its combined active-silicon envelope lies on the LYSO/TS axis. This means $\Delta X = +46.56$ mm and $\Delta Y = -46.08$ mm; ECal height, rotations and internal layer offsets stay unchanged. All HCal positions stay fixed. **Centering increases the rate by 54.2% relative to raising alone**, and by 38.5% relative to the original layout.

Overhead shielding	Raised + centred/hour	Per day	Gain over raised
None	1.949	46.8	+54.2%
3 ft concrete	1.660	39.8	+54.2%

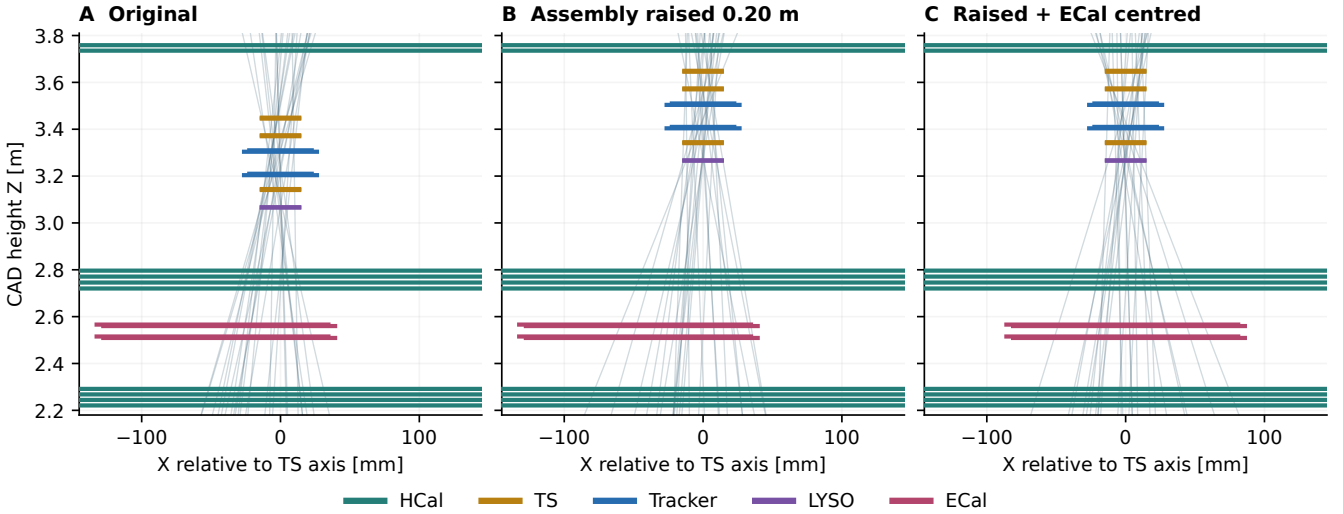


Figure 3. The three layouts with 24 accepted straight rays each, in X – Z . TS, tracker and LYSO move upward together in B and C; ECal is also centred in C. Its Y shift is not visible in this projection. Line thicknesses are enlarged, and HCal extends outside the transverse window.

Checks and limits. Each variation uses eight Sobol scrambles of 2^{18} trials. ROOT agrees with all 4,992 layer decisions in each 192-ray test, including 109 translated active solids for B and 113 for C. Both independent pseudorandom checks agree within 0.16 numerical standard errors. The spectrum and roof model are unchanged. To isolate placement effects, the detector stopping proxy stays at the original 32.70 g cm^{-2} vertical column; it is not remeasured. Scattering, detector inefficiency and mechanical clearances remain unmodelled. These are rate studies, not validated mechanical or transport geometries.

Centering ECal without raising the assembly

Keep TS, tracker, LYSO and all HCal groups at their *original heights*. Move only ECal by $\Delta X = +46.56$ mm and $\Delta Y = -46.08$ mm, using the same active-envelope alignment as on the previous page. ECal height, rotations and internal layer offsets stay unchanged. All 26 active layers are still required.

Centering alone gives 2.163 muons/hour without a roof and 1.842 with 3 ft of concrete. This is a 53.7% increase over the original layout, and 11.0% above the raised-and-centred configuration. In this geometry, raising the assembly is not needed to obtain the gain from ECal alignment.

Removing ECal

The second test physically removes the entire ECal subtree, including its passive materials, while keeping every other subsystem at its original position. The coincidence now requires **22 active layers**: six TS, four tracker, two LYSO and ten HCal. It no longer tests passage through an ECal.

Configuration	Layers	No roof/hour	Concrete/hour	No roof/day
Original	26	1.407	1.199	33.8
ECal centred only	26	2.163	1.842	51.9
ECal removed	22	2.214	1.883	53.1

Here “concrete” means the same 3 ft uniform overhead slab. The 22-layer geometric reference is 1.5730 ± 0.0002 muons/hour for $p > 1$ GeV/c; centering ECal retains 98.5% of that geometrical coincidence rate while still requiring all four ECal layers.

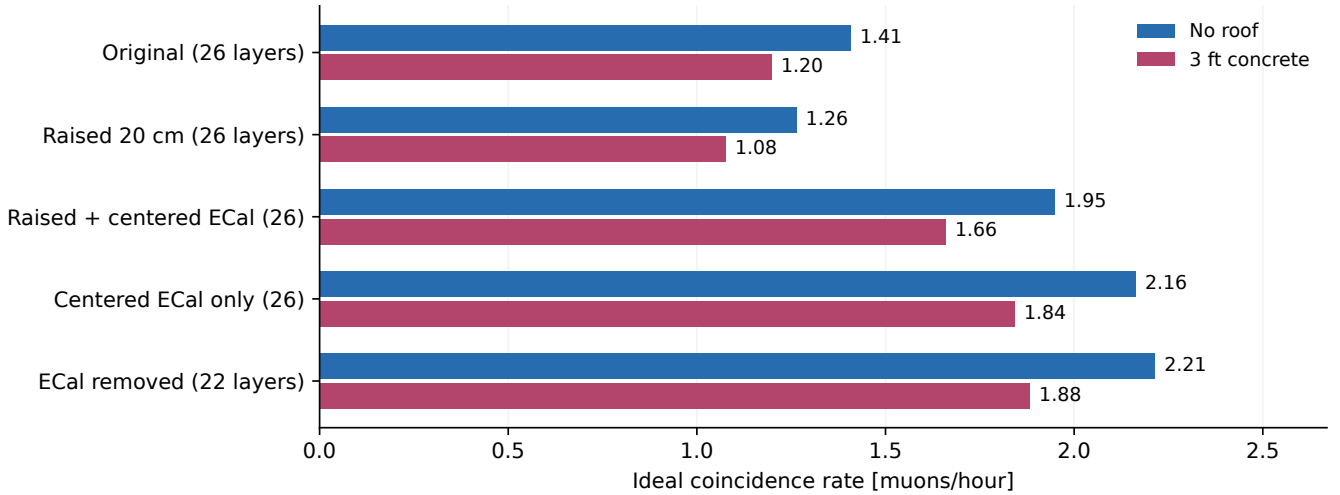


Figure 4. All five configurations under the same incident-spectrum model. The last row uses the relaxed 22-layer requirement and excludes ECal material. The raised rows move TS, tracker and LYSO upward by 200 mm; the two new cases use their original positions.

Material treatment. The centred-only case retains the original 32.70 g cm^{-2} stopping proxy, consistent with the earlier placement studies. For ECal removal, ROOT chords for 96 accepted rays give 24.32 g cm^{-2} for the remaining detector. The same polystyrene CSDA approximation represents that reduced column. At the original column, the 22-layer angular acceptance alone would give 2.195 and 1.869 muons/hour; removing ECal material raises them to the values in the table.

Checks and limits. Eight Sobol scrambles of 2^{18} trials were run for each new case, with independent pseudorandom checks. ROOT agrees with all 4,992 layer decisions for centering and all 4,224 for removal in 192-ray tests. Scattering, straggling, decay, detector inefficiency and mechanical-fit validation remain outside this estimate. The scripts and results are in `ecal_comparison/`; `build_report.sh` rebuilds these sections too.